EBRAHEEM KHALED FARAG

Yale University: Department of Astronomy (YCAA)
(+1) 614 623 4875, ebraheem.farag@yale.edu, ORCID: 0000-0002-5794-4286
Website: debraheem.github.io

RESEARCH INTERESTS

Open-source stellar-evolution software development, validation, and integration testing
Classical pulsations, time-dependent 1D convection, and asteroseismic calibration of stellar interiors
Massive-star evolution, rotation, supernova progenitors, and pair-instability phenomena
Nuclear reaction rates, atomic opacities, and microphysics in stellar models
Stellar neutrino emission and other multimessenger signatures of stellar evolution

POSITIONS

YCAA Prize Fellow *September 2024 - present*
Yale Center for Astronomy & Astrophysics Post-doctoral Research Fellowship Yale University

EDUCATION

Arizona State University *August 2019 - August, 2024 (PhD)*
PhD Astrophysics School of Earth and Space Exploration

The Ohio State University *2014 - 2019 (B.S.)*
Engineering Physics, Mechanical Specialization College of Engineering

TECHNICAL STRENGTHS

Languages	Fortran, Python, LaTeX
Scientific software	MESA, GYRE, Git/GitHub, regression and integration testing
Software leadership	Public releases, module maintenance, test-suite validation, user support

SERVICE

Referee: Astrophysical Journal, Astronomy & Astrophysics, Astronomy and Computing
Scientific and Local Organizing Committees *2026*
Yale/CeNAM Nuclear Asteroseismology Workshop
SESE Graduate Council Recruitment Chair *2022-2023*
Member of Los Alamos National Laboratory Student Association *2015-2023*
Treasurer - The Society of Physics Students 2018-2019

TEACHING

Lecturer: Modules for Experiments in Stellar Astrophysics (MESA) Summer School *2026*
Classical Cepheid Pulsations in MESA-star
TA: Modules for Experiments in Stellar Astrophysics (MESA) Summer School *2018-2025*
TA: AST 113 Introduction to Astronomy Laboratory *2020*

AWARDS

AAS 237 Chambliss Honorable Mention : 2021

Exploring the Black Hole Mass Gap Sensitivity to $^{12}C(\alpha,\gamma)^{16}O$.

PUBLICATIONS

Summary: 20 peer-reviewed papers (1 accepted) and 2 other scholarly publications.
5 first-author and 2 second-author peer-reviewed papers.
459 citations; h-index: 11 (ADS, accessed October 26, 2026).

Peer Reviewed Publications

First Author Publications:

5. "Self-Consistent Nonlinear Classical Cepheid Pulsations During Stellar Evolution with MESA"
   Farag, E., Bellinger, E. P., Mocz, P., Kalici, S., Smolec, R., Kanbur, S., Bettwy, K., & Lindsay, C. 2026, ApJ, 1002, 172

4. "An Expanded Set of Los Alamos OPLIB Tables in MESA: Type-1 Rosseland-mean Opacities and Solar Models"
   Farag, E., Fontes, C. J., Timmes, F. X., Bellinger, E. P., Guzik, J. A., et al. 2024, ApJ, 968, 56

3. "Stellar Neutrino Emission Across The Mass-Metallicity Plane"
   Farag, E., Timmes, F. X., Chidester, M. T., Anandagoda, S., & Hartmann, D. H. 2024, ApJS, 270, 5

2. "Resolving the Peak of the Black Hole Mass Spectrum"
   Farag, E., Renzo, M., Farmer, R., Chidester, M. T., & Timmes, F. X. 2022, ApJ, 937, 112

1. "On Stellar Evolution in a Neutrino Hertzsprung-Russell Diagram"
   Farag, E., Timmes, F. X., Taylor, M., Patton, K. M., & Farmer, R. 2020, ApJ, 893, 133; Erratum: 2024, ApJ, 965, 192

Contributing Author Publications:

15. "A Grid of Fast-rotating, Chemically Homogeneous Supernova and/or Long-GRB Progenitors" **(Accepted)**
    Renzo, M., Gottlieb, O., Chan, H. S., Goldberg, J. A., Grichener, A., Sen, K., Shah, N., **Farag, E.**, & Cantiello, M. 2026, ApJS, in press; arXiv:2606.21824

14. "Benchmarking Machine Learning Emulators of Stellar Evolution for Precision Asteroseismology"
    Gluck, N., Bellinger, E. P., Liang, Y., **Farag, E.**, Saunders, N., Kalici, S., Lindsay, C. J., & Basu, S. 2026, ApJ, 1010, 33

13. "Red-Giant Asteroseismology of Low-Mass Population III Stars"
    Ferreira, T., Bellinger, E. P., **Farag, E.**, & Lindsay, C. J. 2026, ApJ, 1001, 41

12. "Evolution of Low-Mass Population III Stars: Convection, Mass Loss, Nucleosynthesis, and Neutrinos"
    Ferreira, T., Bellinger, E. P., **Farag, E.**, & Lindsay, C. J. 2026, ApJ, 1000, 147

11. "Evolutionary Tracks and Spectral Properties of Quasi-stars and Their Correlation with Little Red Dots"
    Santarelli, A. D., **Farag, E.**, Bellinger, E. P., Natarajan, P., Naidu, R. P., Campbell, C. B., & Caplan, M. E. 2026, ApJL, 998, L4

10. "MESA-QUEST: Tracing the Formation of Direct Collapse Black Hole Seeds via Quasi-stars"
    Santarelli, A. D., Campbell, C. B., **Farag, E.**, Bellinger, E. P., Natarajan, P., & Caplan, M. E. 2026, ApJ, 998, 150

9. "Massive Stellar Cannibals: How Stellar Mergers Drive Mass Loss in Extremely Massive Stars"
   Roman-Garza, J., Fragos, T., Charbonnel, C., Ramírez-Galeano, L., Kruckow, M., & **Farag, E.** 2026, A&A, 707, A163

8. "Solar-like Oscillations in Accreting Pre-main-sequence Stars: Insights and Prospects"
Jorgensen, J., Zwintz, K., **Farag, E.**, Vorobyov, E. I., & Steindl, T. 2025, ApJ, 995, 223

7. "Nuclear Neural Networks: Emulating Late Burning Stages in Core-collapse Supernova Progenitors"
Grichener, A., Renzo, M., Kerzendorf, W. E., Farmer, R., de Mink, S. E., Bellinger, E. P., Chan, C. K., Chen, N., **Farag, E.**, & Justham, S. 2025, ApJS, 279, 49

6. "Photons from Neutrinos: The Gamma-Ray Echo of a Supernova Neutrino Burst"
Lunardini, C., Loeffler, J., Mukhopadhyay, M., Hurley, M. J., **Farag, E.**, & Timmes, F. X. 2024, ApJ, 969, 149

5. "Seismic Signatures of the $^{12}C(\alpha, \gamma)^{16}O$ Reaction Rate in White Dwarf Models with Overshooting"
Chidester, M. T., Timmes, F. X., & **Farag, E.** 2023, ApJ, 954, 51

4. "JWST Imaging of Earendel, the Extremely Magnified Star at Redshift z = 6.2"
Welch, B., Coe, D., Zackrisson, E., de Mink, S. E., Ravindranath, S., Anderson, J., Brammer, G., Bradley, L., Yoon, J., Kelly, P., Diego, J. M., Windhorst, R., Zitrin, A., Dimauro, P., Jimenez-Teja, Y., Abdurro'uf, Nonino, M., Acebron, A., Andrade-Santos, F., Avila, R. J., Bayliss, M. B., Benitez, A., Broadhurst, T., Bhatawdekar, R., Bradac, M., Caminha, G. B., Chen, W., Eldridge, J., **Farag, E.**, Florian, M., Frye, B., et al. 2022, ApJL, 940, L1

3. "On Trapped Modes in Variable White Dwarfs as Probes of the $^{12}C(\alpha, \gamma)^{16}O$ Reaction Rate"
Chidester, M. T., **Farag, E.**, & Timmes, F. X. 2022, ApJ, 935, 21

2. "Observing Intermediate-mass Black Holes and the Upper Stellar-mass Gap with LIGO and Virgo"
Mehta, A. K., Buonanno, A., Gair, J., Miller, M. C., **Farag, E.**, deBoer, R. J., Wiescher, M., & Timmes, F. X. 2022, ApJ, 924, 39

1. "On the Impact of ^{22}Ne on the Pulsation Periods of Carbon-Oxygen White Dwarfs with Helium-dominated Atmospheres"
Chidester, M. T., Timmes, F. X., Schwab, J., Townsend, R. H. D., **Farag, E.**, Thoul, A., Fields, C. E., Bauer, E. B., & Montgomery, M. H. 2021, ApJ, 910, 24

Other Scholarly Publications

2. "Investigating Opacity Modifications and Reaction Rate Uncertainties to Resolve the Cepheid Mass Discrepancy"
Guzik, J. A., **Farag, E.**, Ostrowski, J., Evans, N. R., Neilson, H., Moschou, S., & Drake, J. J. 2021, in Astronomical Society of the Pacific Conference Series, Vol. 529, RR Lyrae/Cepheid 2019: Frontiers of Classical Pulsators, ed. K. Kinemuchi, C. Lovekin, H. Neilson, & K. Vivas, 79

1. "Sound speed and oscillation frequencies for solar models evolved with Los Alamos ATOMIC opacities"
Guzik, J. A., Fontes, C. J., Walczak, P., Wood, S. R., Mussack, K., & **Farag, E.** 2016, IAU Focus Meeting, 29B, 532–535

SOFTWARE

3. "MESA: Modules for Experiments in Stellar Astrophysics"
Maintainer and developer 2023–present

2. "pyfortmesa"
Farag, E. 2026, software, version 0.6.0

1. "MESA-QUEST: MESA Quasi-star Evolutionary Simulation Toolkit"
Santarelli, A. D., Campbell, C. B., **Farag, E.**, Bellinger, E. P., Natarajan, P., & Caplan, M. E. 2026, Astrophysics Source Code Library, ascl:2604.006

PRESENTATIONS & INVITED TALKS

18. Nuclear Physics Seminar (title TBA) *Invited Seminar (scheduled)*
University of Notre Dame, Notre Dame, IN; November 23, 2026

17. Nuclear Reactions in MESA *Invited Keynote*
Yale/CeNAM Nuclear Asteroseismology Workshop, New Haven, CT, 2026

16. Classical Cepheid Pulsations in MESA-star *Invited Lecturer*
MESA Summer School, Jackson, WY, 2026

15. From Stellar Evolution to Pulsation: Post-Main Sequence Massive Star Pulsations *Invited Seminar*
International Space Science Institute, Bern, Switzerland, 2026

14. Self-consistent Non-linear Radial Pulsations during Stellar Evolution *Invited Seminar*
Max Planck Institute for Astrophysics, 2025

13. Multi-Messenger Signals from Stellar Evolution *Invited Colloquium*
Yale Astronomy Colloquium, 2025

12. On Stellar Neutrino Emission Across the Mass-Metallicity Plane *Contributed Presentation*
Farag, E., Timmes, F., & Chidester, M. 2024, Bulletin of the American Astronomical Society, 56(2), 116.06

11. Neutrino Emission from Stars *Contributed Poster*
Farag, E., Chidester, M., & Timmes, F. 2023, in AAS/High Energy Astrophysics Division, Vol. 55, AAS/High Energy Astrophysics Division, 115.20

10. Resolving The Peak Of The Black Hole Mass Spectrum *Contributed Poster*
Farag, E., Renzo, M., Farmer, R., Chidester, M., & Timmes, F. 2023, in AAS/High Energy Astrophysics Division, Vol. 55, AAS/High Energy Astrophysics Division, 116.62

9. Resolving The Peak Of The Black Hole Mass Spectrum *Contributed Talk*
Farag, E., Renzo, M., Farmer, R., Chidester, M., & Timmes, F. 2023, in American Astronomical Society Meeting Abstracts, Vol. 55, American Astronomical Society Meeting Abstracts, 259.04

8. Recurrent Novae and Steady Burning *Invited Lecturer*
(MESA) Summer School, Santa Barbara, CA, 2022

7. Resolving The Peak of The Black Hole Mass Spectrum *Invited Seminar*
Farag, E. K., Renzo, M., Farmer, R., Chidester, M., & Timmes, F. 2022
Los Alamos Distinguished Seminar Series, Los Alamos, NM

6. On Stellar Evolution in a Neutrino Hertzsprung-Russell Diagram *Contributed Flash Talk*
XIX International Workshop on Neutrino Telescopes, online; February 19, 2021

5. Exploring The Black hole Mass Gap With Revised Nuclear Reaction Rates *Contributed Poster*
Farag, E., & Timmes, F. 2021, in American Astronomical Society Meeting Abstracts, Vol. 53, American Astronomical Society Meeting Abstracts, 524.03

4. Modeling Cepheid Variable Stars Using the Open-Source MESA Code *Coauthored Abstract*
Guzik, J. A., **Farag, E.**, Ostrowski, J., et al. 2020, The Journal of the American Association of Variable Star Observers, vol. 48, no. 1, p. 102

3. Modeling Pulsations of Cepheid Variables using the Open-Source MESA Code *Coauthored Talk*
   Guzik, J. A., **Farag, E.**, Ostrowski, J., et al. 2020, in The 35th Annual New Mexico Symposium, ed. A. D. Kapinska, 5; presented by J. A. Guzik

2. Is There A Solar Model Solution to The Faint Young Sun Paradox? *Invited Seminar*
   Farag, E. K. 2019
   Los Alamos Distinguished Seminar Series, Los Alamos, NM

1. A diffusive radiation hydrodynamics code, xRage, is implemented to compare radiation flow with experimental data from the Omega laser facility *Coauthored Poster Abstract*
   VanDervort, R., Elgin, L., **Farag, E.**, et al. 2015, Bulletin of the American Physical Society, 60(19), UP12.00023

ACCEPTED PROPOSALS

1. A Strongly Magnified Individual Star and Parsec-Scale Clusters Observed in the First Billion Years at z = 6, JWST Proposal. Cycle 1, ID. #2282
   Coe, D., Acebron, A., Avila, R., et al. 2021,

2. OzGrav: The Australian Research Council Centre of Excellence for Gravitational Wave Discovery
   For the MESA Down Under Winter School, hosted by the University of Sydney, 2024